Guilhermino Lucas

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Summary

Forward Deployed AI Engineer profile: full-stack engineer who ships production generative-AI and agentic systems end to end, from Python/FastAPI backends and LangGraph/LangChain agent flows to TypeScript/React frontends. Three years of AI and data engineering work (since November 2023) building advanced RAG platforms and MCP (Model Context Protocol) tool ecosystems adopted by real clients. Production systems, not prototypes: a criminal-background and lawsuit analysis solution that cut analysis time by up to 90% (from nearly a manual day to under 30 seconds), AI-generated assembly minutes adopted by 135 management companies, and a support knowledge base that cut response time by up to 80%. Drives AI-first engineering across teams with custom Claude Code agents and skills that raise the quality and consistency of AI-generated code. Currently studying Data Science and Artificial Intelligence at PUC Minas.

Experience

Group Software · AI Engineer

June 2025 – Present

- Built and shipped Group Verifica end to end (frontend, backend, AI, and database) for the condominium and HR markets: a LangGraph-powered risk engine with tools querying real-time background-check APIs, configurable in 3 risk levels with full decision traceability, cutting criminal-background and lawsuit analysis from nearly 1 manual day to **under 30 seconds (~90% reduction)**, reaching 53 clients within 20 days, 72% purchase interest, and **R$4,750 MRR in the first 45 days**
- Architected a production RAG platform (two-layer ingestion, overlapping chunking, semantic search, re-ranking) for condo documents, consulted by residents via WhatsApp and site/app and adopted by support as the central knowledge base, delivering **up to 80% support response-time reduction**
- Shipped AI transcription with auto-generation of assembly minutes plus a transcript-mining lead-generation engine: **482 minutes generated in 49 days** (68.1% organic adoption, peak 77.3%), eliminating **1,446 specialist hours across 135 management companies**, and 239 cross-sell leads in 78 days at ~R$107 total AI cost (~93% below market platforms)
- Co-architected GIA, the central AI platform orchestrating multiple MCP servers (~20 AI tools across 3 products, dynamic routing by client entitlement and conversational context, JWT per-product auth, document chat via RAG over pgvector, multi-tenant plug-and-play layer), and built the **production MCP tool set (19 tools with strict Zod structured outputs)** with two-layer authentication, consistent across stdio and HTTP transports
- Ran LLMOps for the AI platform: multi-model orchestration via OpenRouter with systematic model comparison (GPT-5-mini, GPT-4.1-mini, Claude 3 Haiku, Gemini, Gemma), per-conversation, per-client, and per-model **cost auditing**, automated red teaming with promptfoo (client isolation, auth bypass, instruction injection), and end-to-end LLM observability (OpenTelemetry to Datadog, self-hosted Langfuse)
- Worked AI-first daily: used Claude Code, Cursor, GitHub Copilot, and Codex across design, implementation, testing, and code review, and **built custom Claude Code agents and skills for the development team**, raising the quality of AI-generated code and keeping it consistent across everyone working on the project
- Upheld the engineering bar behind these systems: **2,870 automated tests with a 90% coverage gate in CI**, hexagonal architecture with LangGraph/LangChain confined to the infrastructure layer, deployed to Azure and 3 AWS ECS environments via Terraform and Docker

Tech stack: Python, FastAPI, TypeScript, React, LangChain, LangGraph, MCP (Model Context Protocol), OpenRouter, OpenAI GPT-4o, Azure OpenAI, Claude API, pgvector, Qdrant, Azure (containers + managed PostgreSQL), AWS ECS, Terraform, Docker, OpenTelemetry, Datadog, Langfuse, promptfoo, Zod

H&P · Data Scientist / AI Engineer

November 2023 – June 2025

- Redesigned the relational database structure with software-engineering best practices: 90 tables redesigned with a 70% performance increase and an estimated ~60% saving in database-maintenance time, preparing the infrastructure to support AI workflows

- Pioneered the deployment of generative-AI solutions across the company, from identified strategic use cases, delivering up to 80% reduction in report-writing time for the responsible team
- Ran internal AI study groups on RAG, chunks, embeddings, and LLM selection: 12 professionals upskilled during the program
- Owned governance and integrity of the Fundão dam breach (Mariana/MG) registration databases: absolute data accuracy for strategic decisions and high-impact reports, serving 1,200 people

Tech stack: Relational database architecture and governance; generative-AI (LLM) solutions; RAG, chunking, embeddings, and LLM selection (internal AI study program)

The Brooklyn Brothers · Mid-Level Front-End Developer

March 2022 – January 2023

- Developed and improved high-performance web interfaces for Rexona, one of the top clients in the global portfolio: 23% SEO improvement plus better user experience, operational efficiency, and accessibility; ~30% performance improvement
- Worked hand-in-hand with development teams in India and the United Kingdom in a global team of 15 people, in a multilingual, agile (Scrum) environment

Tech stack: React, Context API, JavaScript (ES6+), HTML5, CSS3, responsive web components

Kenzie Academy Brasil · Peer Coach (Teaching Assistant)

August 2021 – February 2022

- Coached 100 students on React, CSS, and HTML (Virtual DOM, styled-components) and versioning with Git/GitHub, who raised the quality of their projects

Tech stack: React, CSS, HTML, Git/GitHub, Context API, Redux

Red Lion Inglês Britânico · English Instructor

March 2020 – May 2020

- Taught English to students of all age groups, creating didactic materials and interactive activities and tracking progress with tests and constructive feedback

Core Skills

- **GenAI (Generative AI) and LLMs:** OpenAI, OpenRouter, Azure OpenAI, Claude API, GPT, Gemini, Anthropic, prompt engineering, LLM deployment, speech-to-text, text-to-speech, voice AI
- **Agentic AI, Agents and RAG:** AI agents, agentic AI, LangChain, LangGraph, advanced RAG, multi-agent systems, agent tool integration, tool/function calling, MCP (Model Context Protocol), semantic search, context engineering, embeddings, vector search, vector databases (pgvector, Qdrant, Chroma, Pinecone)
- **AI Tooling and Ops:** Claude Code, Cursor, GitHub Copilot, Codex, custom Claude Code agents and skills, AI evaluations, LLMOps practices (automated red teaming, LLM observability, per-conversation LLM cost auditing), AI observability, CI/CD, Docker, Terraform, Azure, Azure AI Foundry, Vertex AI, pytest
- **Languages and Frameworks:** Python, TypeScript, JavaScript, FastAPI, React, SQLAlchemy, Node.js, Vitest
- **Data and Machine Learning:** PostgreSQL, MongoDB, Redis, data engineering, API development, pandas, Power BI, Machine Learning fundamentals (LLM evaluation and model selection)
- **Ways of Working:** Agile methodologies, Scrum, automated testing, multi-tenancy, hexagonal architecture

Certifications

Model Context Protocol (MCP) Professional Certification · Anthropic (certificate of completion), 2026

Additional Anthropic certifications: MCP: Advanced Topics · Introduction to MCP · Introduction to Subagents · Introduction to Agent Skills

Languages

Portuguese (native)

English (full professional)